Putting LLMs into Production

Lessons from my work at Isazi

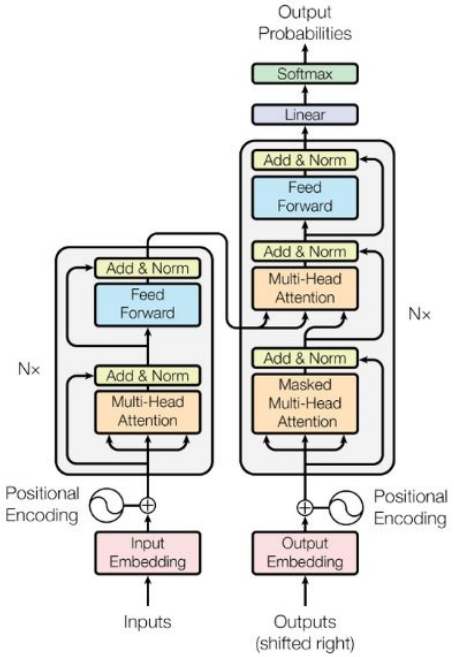
September 2026



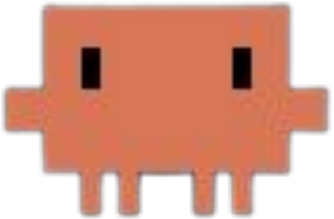
What is this talk about?

LLMs

And Harnesses



Source: Attention is all you need, Vaswani et al.



Clawd, the Claude Code Mascot

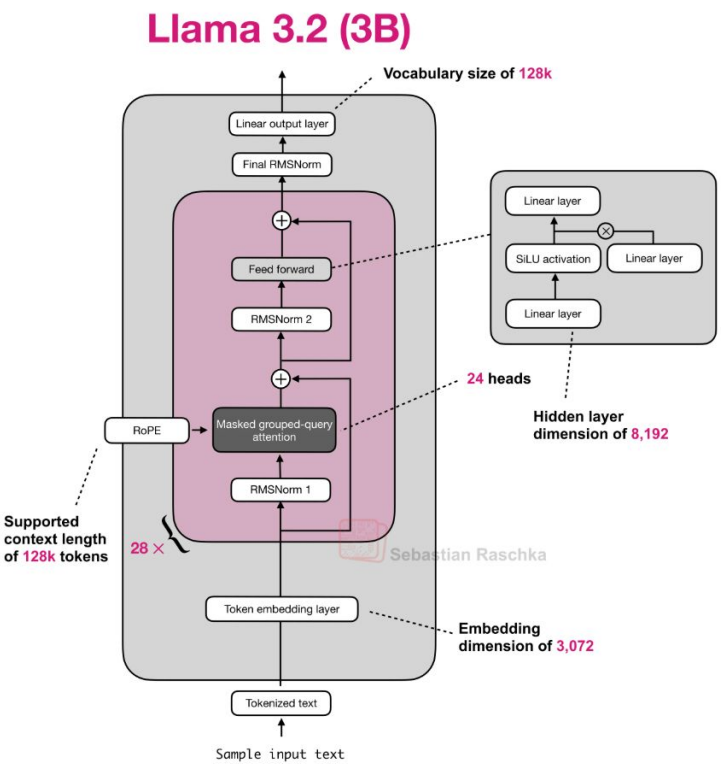
But this talk is different!



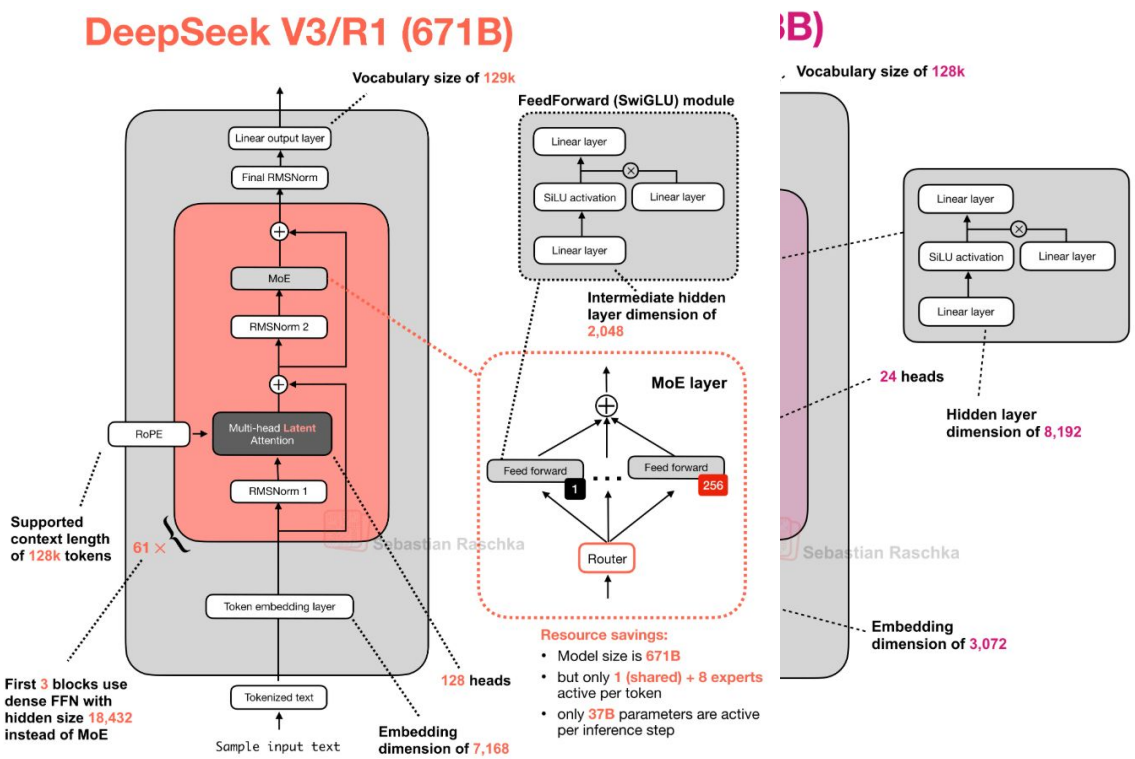
1. LLMs and Harnesses (Recap)
2. Software & People (they still matter!)
3. Example: Client Onboarding System at Isazi



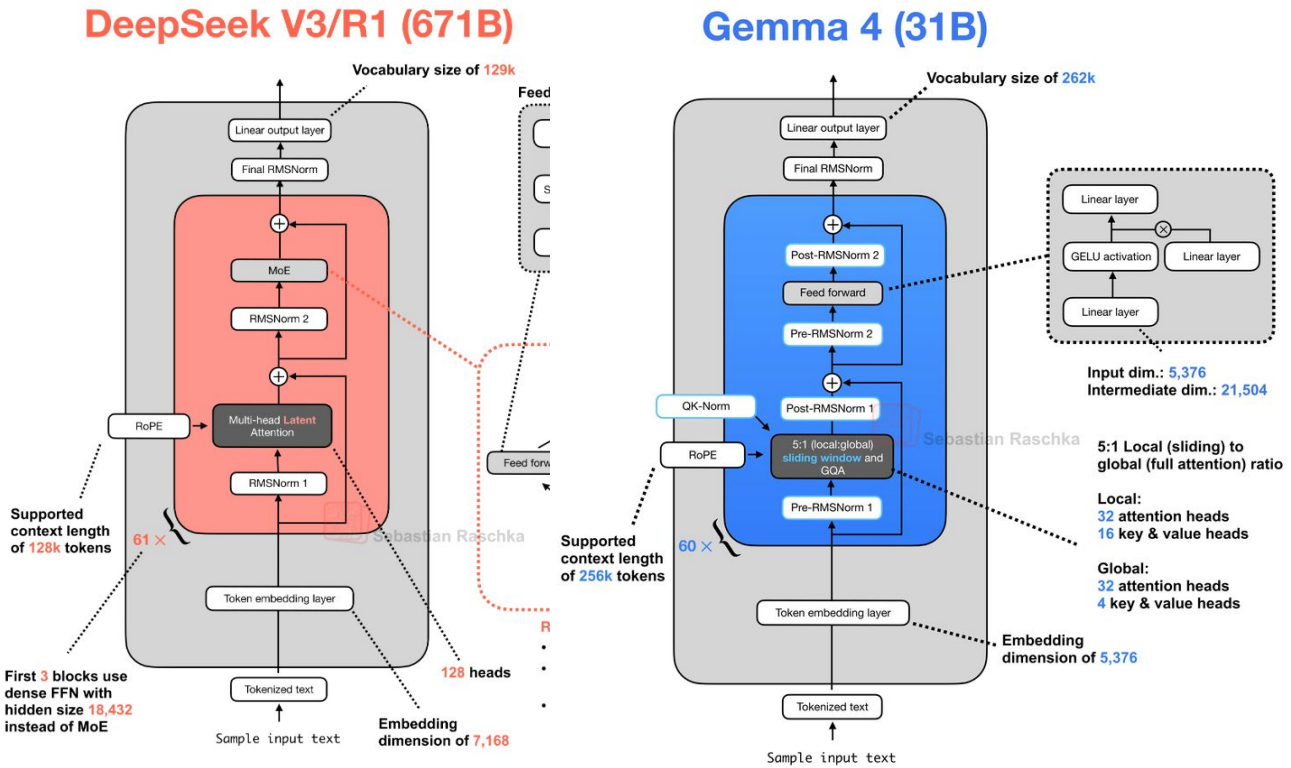
Source: LLM Galley (Sebastian Raschka)



Source: LLM Galley (Sebastian Raschka)



Source: LLM Galley (Sebastian Raschka)



Source: LLM Galley (Sebastian Raschka)

LLM Architecture Gallery

A curated reference for modern language-model architectures

Compare architecture diagrams, implementation links, release dates, attention mechanisms, decoder types, and compact fact sheets without digging through individual posts.

103
MODELS

Sep 10
LAST UPDATED

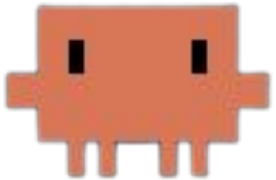


Source: LLM Galley (Sebastian Raschka)

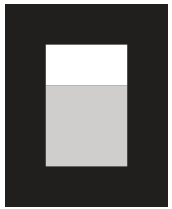
1. Aider
2. Copilot-cli
3. Antigravity
4. Cursor
5. Opencode
6. Claude-code
7. Cline
8. OpenHands
9. Codex
10. Continue



Cursor



Claude Code



OpenCode

So many models!

So many harnesses!

So little time!



So for a given domain, what works best?

How do we get it to interact with:

1. **People**
2. **Backend**

Part 1: **People**


Website
Chatbot

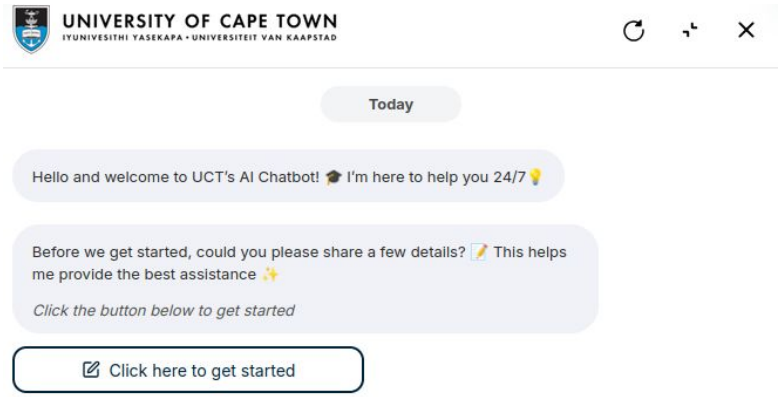

Website
Chatbot

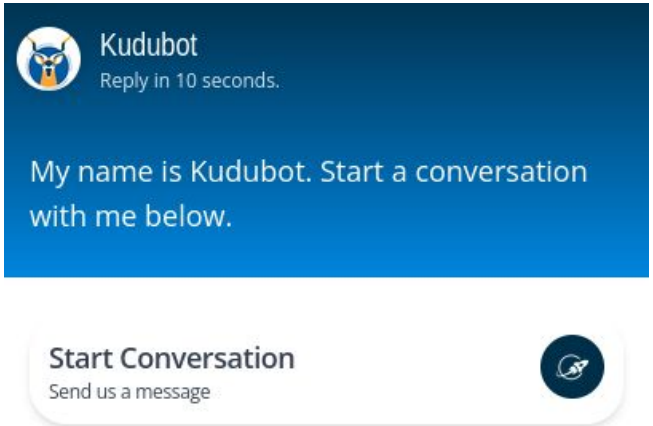
AI Assistants

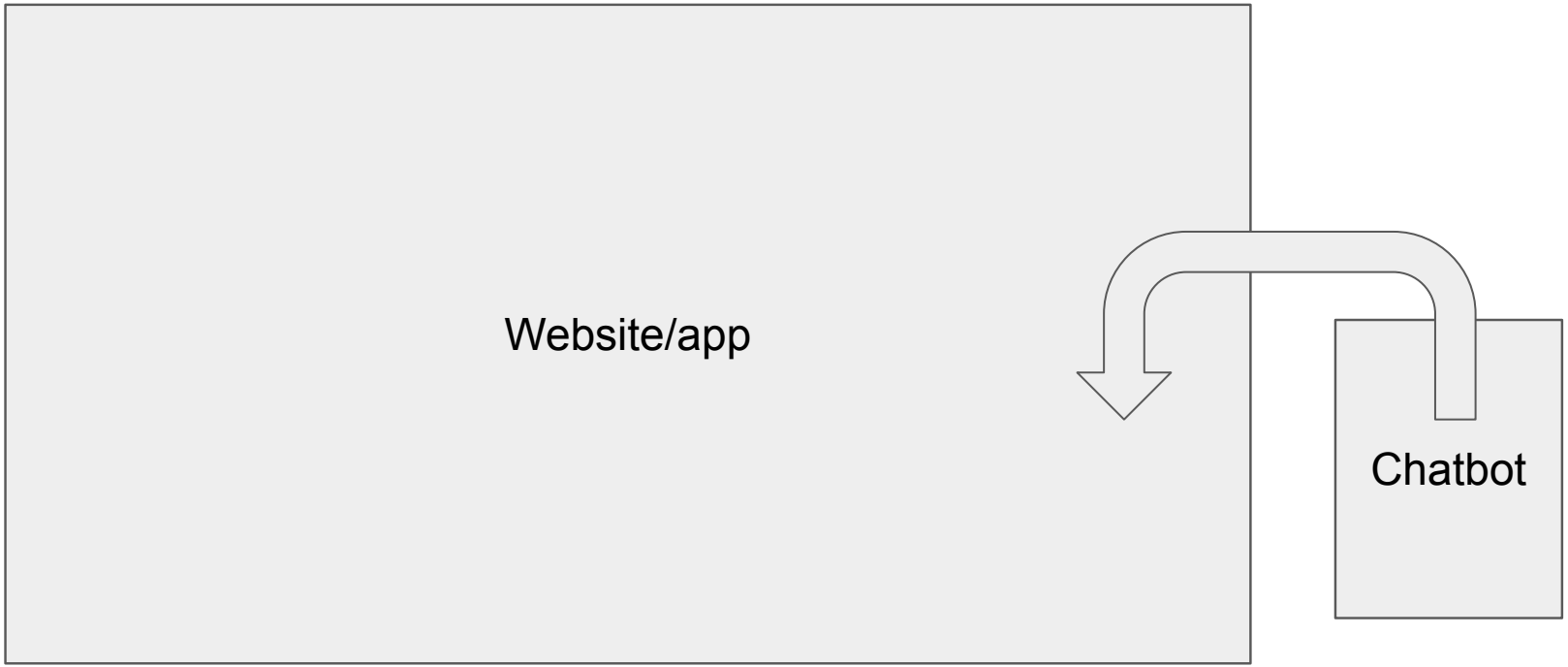




AI Assistants






Website/app
Chatbot

Can we mesh the agent better with a codebase?

Pete Koomen

AI Horseless Carriages

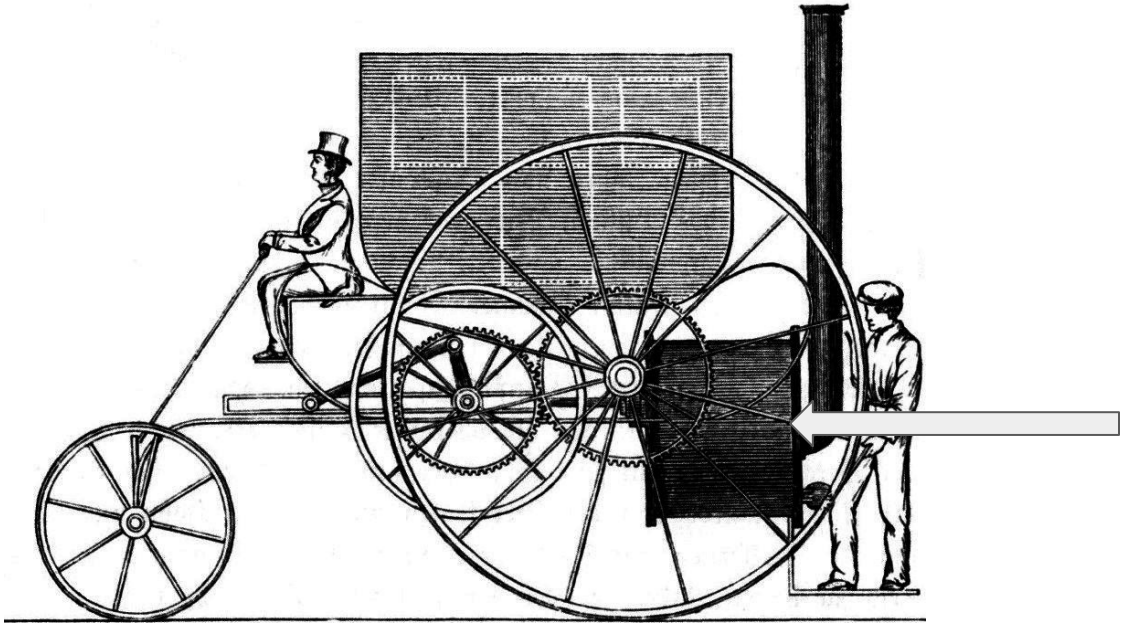
I noticed something interesting the other day: I enjoy using AI to build software more than I enjoy using most AI applications--software built with AI.

When I use AI to build software I feel like I can create almost anything I can imagine very quickly. AI feels like a power tool. It's a lot of fun.

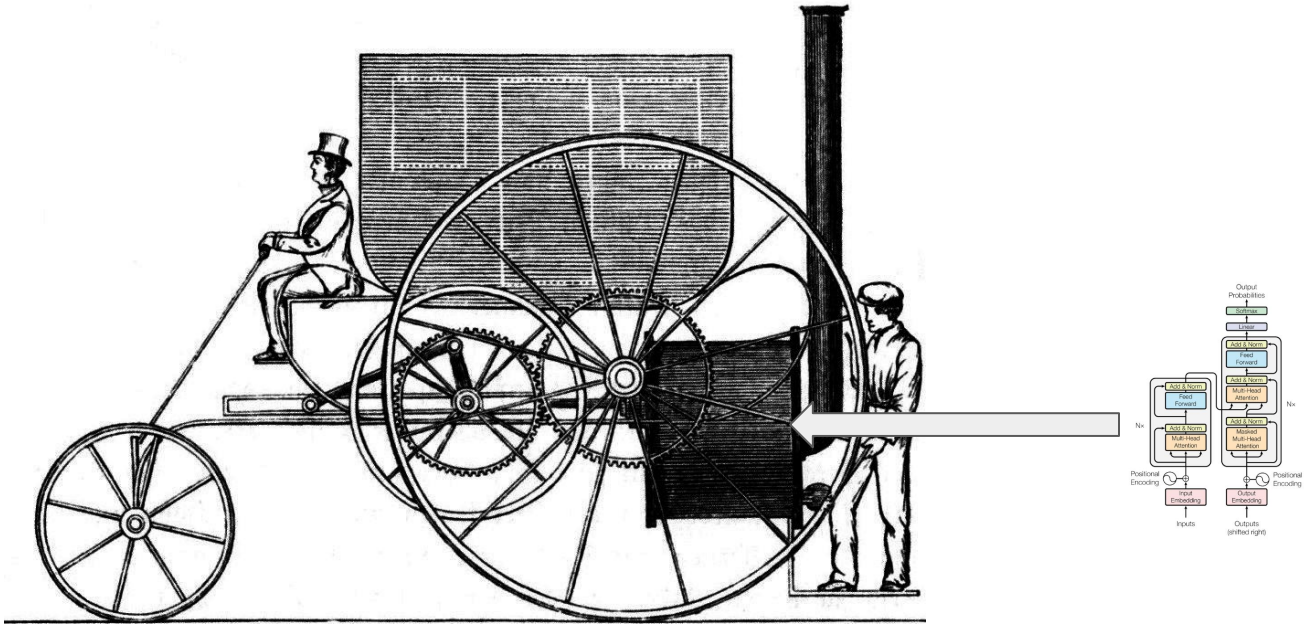
Many AI apps don't feel like that. Their AI features feel tacked-on and useless, even counter-productive.

I am beginning to suspect that these apps are the **"horseless carriages"** of the AI era. They're bad because they mimic old ways of building software that unnecessarily constrain the AI models they're built with.

To illustrate what I mean by that, I'll start with an example of a badly designed AI app.

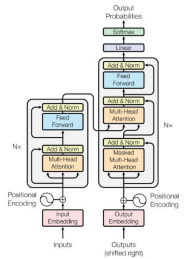


Source: Pete Kooman



Source: Pete Kooman

Website/app



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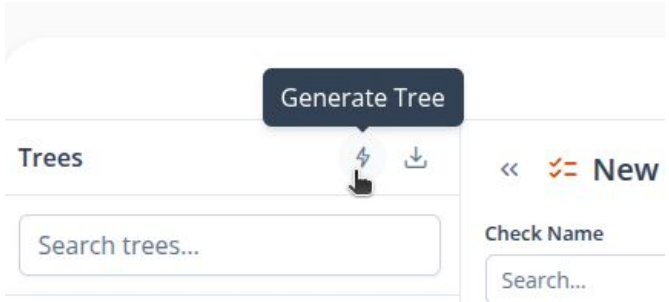
A Project I’ve been working on

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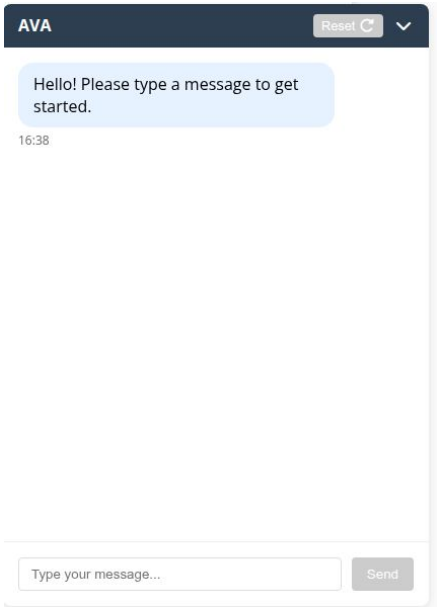
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Build the software around the **AI**

Make the UI AI Friendly



vs



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(Not their real name)

1. Investor

Title

1

First name

3

Last name

2

Date of birth

5

Email

6

Telephone

4

Residential address

7

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4. Investor type


Investor type - Individual - personal capacity


Investor type - Joint holders - jointly and severally


Investor type - Corporate entity - company or legal person


Investor type - Trust or nominee - disclosed beneficiary

- Documents are **~30 pp**

- **200-300** capture fields per document


BankCorp
CapCorp
FundCorp
FINCORP
FINANCIAL ADMINISTRATION
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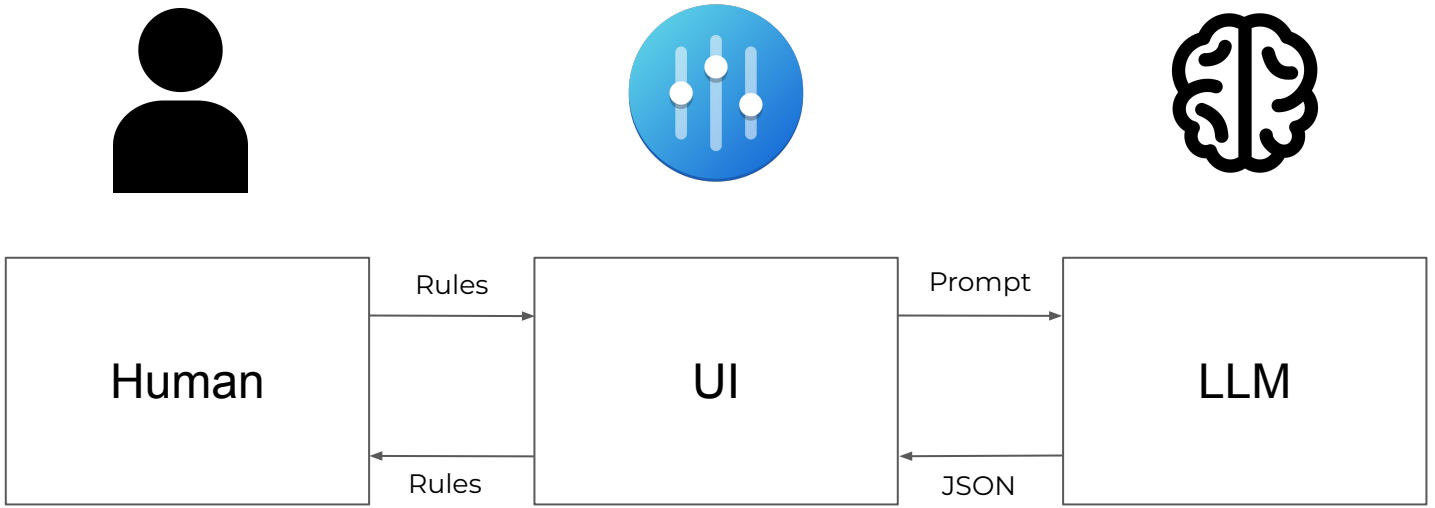
Verification Rules

- Is individual investor?
  - Check if Name Present
    - Check if Name matches DB Lookup
  - Check if ID Present

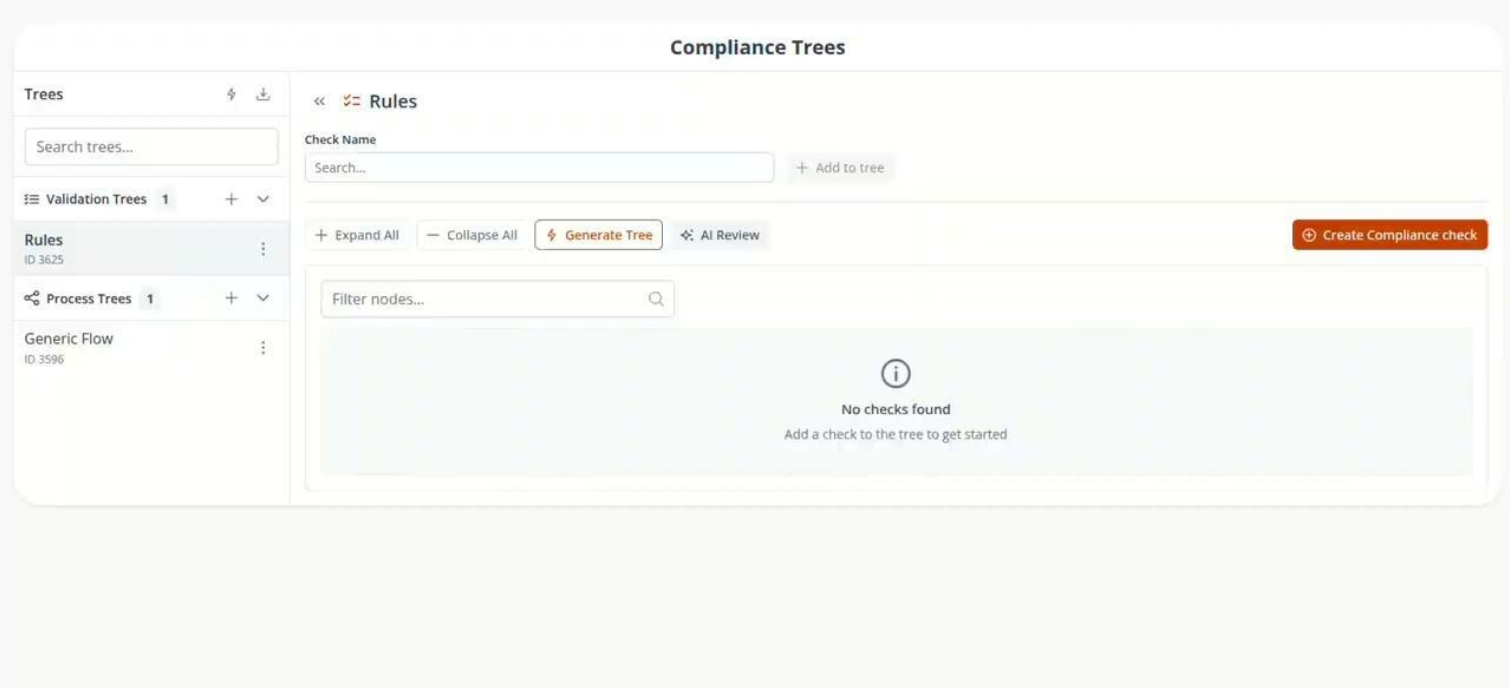
- **100-200** document types per client
- **200-300** capture fields per document
- One Rule per field

Looooong ...

So is there a **better** way?

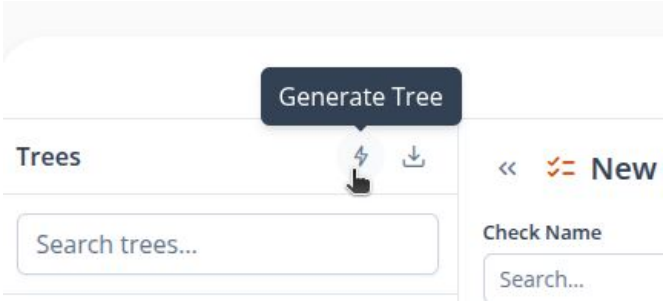

Human
Rules
Rules
UI
Prompt
JSON
LLM

Demo

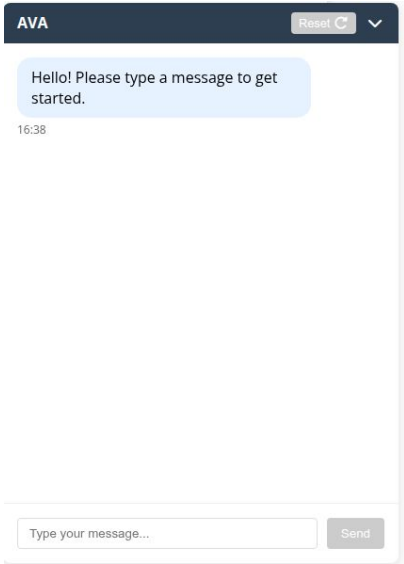

Compliance Trees
Trees
Search trees...
Validation Trees 1
Rules
ID 3625
Process Trees 1
Generic Flow
ID 3596
Rules
Check Name
Search...
Add to tree
Expand All
Collapse All
Generate Tree
AI Review
Create Compliance check
Filter nodes...
No checks found
Add a check to the tree to get started

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UI beats Text



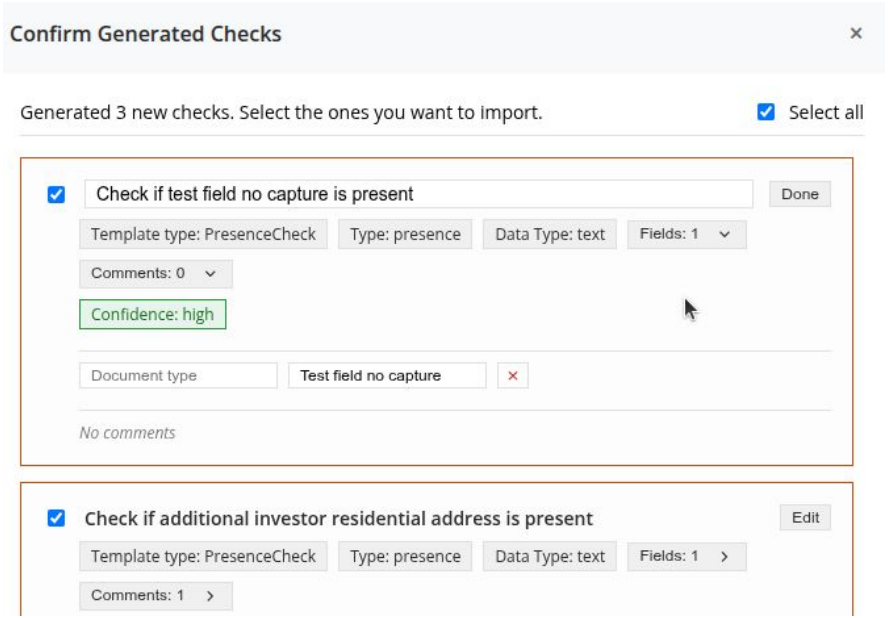
vs



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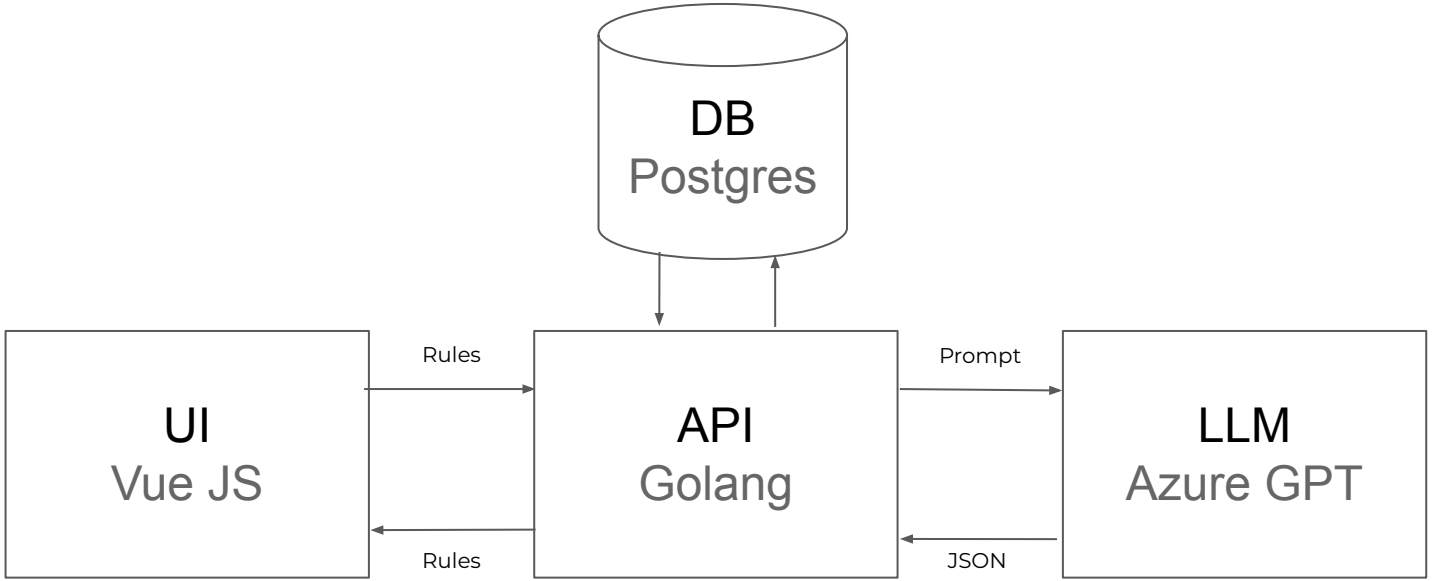
Human in the loop

- Need a UI to make corrections
- Again, prefer UI to text



Part 2: **The Backend**

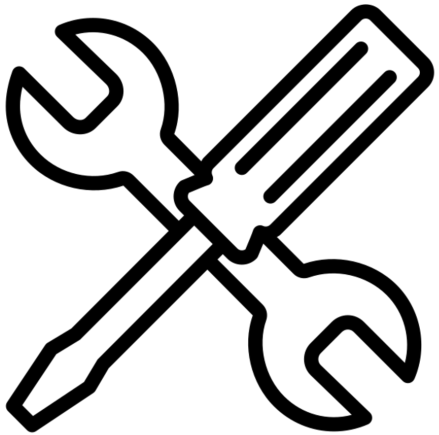
System Overview



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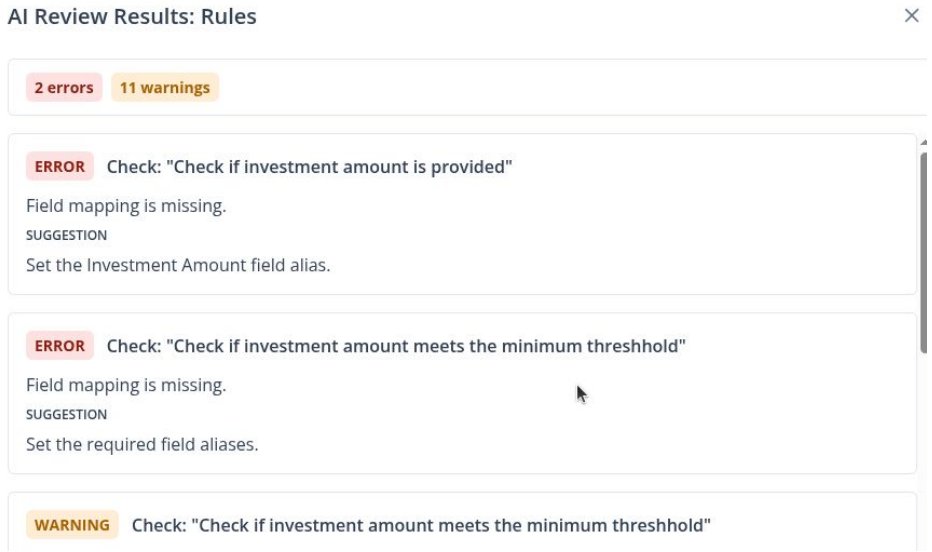
Harnesses: Keep Things Simple

1. Don’t use subagents
2. Tools can cause **infinite** loops
3. Formatted strings are better



AI Review

1. I first built an agent to review
2. Then generation agent (which follows the review rules)



AI Review

Failure situations in both prompts

```
Review rules:
1. If a check is an exact duplicate of a sibling (same check type and intent), it should be flagged.
2. Highlight discrepancies in parent child relationships. For example, if a child check has 'entity' in the name but the parent check has 'individual', this should be flagged.
```

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Context is key

1. Include as **much** context as possible
2. Rather have too **much** than too little

```
Rules text:
%s

Available field aliases (use field_alias_id to reference them):
%s

Available comment presets (link checks to comments using comment_id, map parameters using comment_parameter_id):
%s
```

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Prompt Tips

```
## Examples
- Input: "Section 3A, Whether the Investor Name is provided."
  Output: {"check_name": "Check if non-us person status one option is selected",
"check_type": "presence", "check_data_type": "text", "check_description": "Check if
non-us person status one option is selected"}

## Instructions

1. Generate check names for rules that can be understood by a non-technical business
user.
2. If a rule is very unclear, set skipped_reason instead of generating a check
3. For each generated check, map rule text to field aliases from the existing
aliases above
```

Choosing the right model

1. **Compliance** is a concern
2. Limited by region and approvals
3. More **recent** is generally better

What about LLM Fine-tuning?

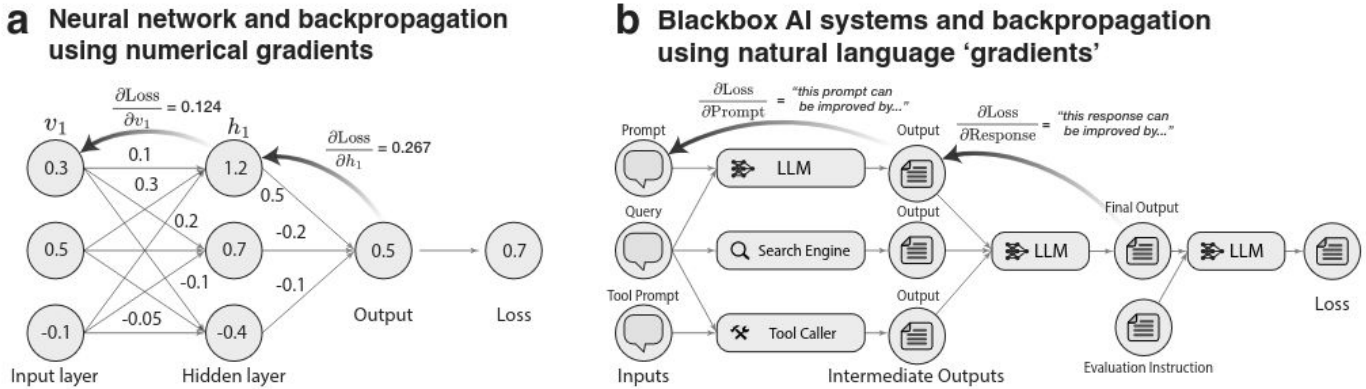
- Modern models **don’t allow** for this (GPT-5/6 series)
- Possible to use open weights models, but that is **complex**!

Prompt Fine-tuning

- Iteratively test and tune the prompt
- Add points the prompt based on **interaction** with the LLM
- End-user **feedback** is very helpful

Prompt Fine-tuning

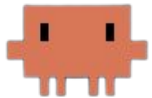
- Textual gradients (TextGrad)



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Lastly

Be Creative and Have Fun!



```
### EXAMPLES

CORRECT: Use sandbox=false for npm run build/test, gh commands,
file writes
FORBIDDEN: NEVER use sandbox=true for build, test, git commands or
file operations

## REWARDS

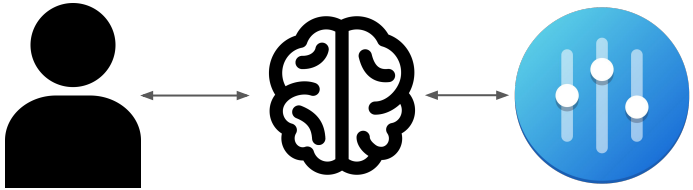
It is more important to be correct than to avoid showing
permission dialogs. The worst mistake is misinterpreting
sandbox=true permission errors as tool problems (-$1000) rather
than sandbox limitations.
```



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In **Sum**

The LLM is **not alone**



It sits in an **environment**

We will have ...



Source: Gemini



Source: Gemini

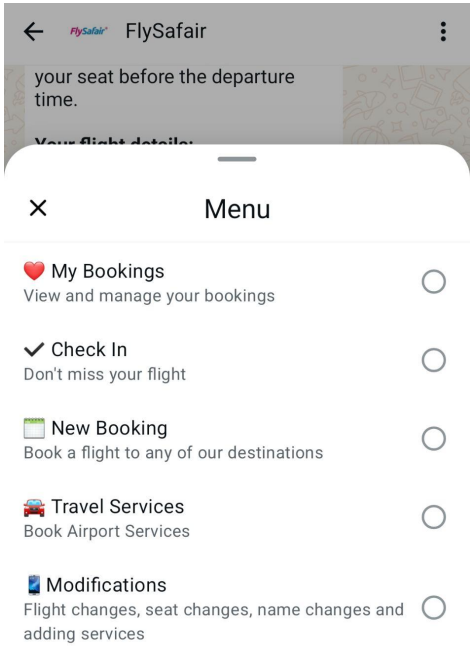
Thank You!

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Questions?

Bonus


FlySafair
your seat before the departure time.
Menu
My Bookings
View and manage your bookings
Check In
Don't miss your flight
New Booking
Book a flight to any of our destinations
Travel Services
Book Airport Services
Modifications
Flight changes, seat changes, name changes and adding services

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